

Epidemiologic Investigation of Immune-Mediated Polyradiculoneuropathy among Abattoir Workers Exposed to Porcine Brain

Background In October 2007, a cluster of patients experiencing a novel polyradiculoneuropathy was identified at a pork abattoir (Plant A). Patients worked in the primary carcass processing area (warm room); the majority processed severed heads (head-table). An investigation was initiated to determine risk factors for illness.

Methods and Results Symptoms of the reported patients were unlike previously described occupational associated illnesses. A case-control study was conducted at Plant A. A case was defined as evidence of symptoms of peripheral neuropathy and compatible electrodiagnostic testing in a pork abattoir worker. Two control groups were used - randomly selected non-ill warm-room workers (n = 49), and all non-ill head-table workers (n = 56). Consenting cases and controls were interviewed and blood and throat swabs were collected. The 26 largest U.S. pork abattoirs were surveyed to identify additional cases. Fifteen cases were identified at Plant A; illness onsets occurred during May 2004–November 2007. Median age was 32 years (range, 21–55 years). Cases were more likely than warm-room controls to have ever worked at the head-table (adjusted odds ratio [AOR], 6.6; 95% confidence interval [CI], 1.6–26.7), removed brains or removed muscle from the backs of heads (AOR, 10.3; 95% CI, 1.5–68.5), and worked within 0–10 feet of the brain removal operation (AOR, 9.9; 95% CI, 1.2–80.0). Associations remained when comparing head-table cases and head-table controls. Workers removed brains by using compressed air that liquefied brain and generated aerosolized droplets, exposing themselves and nearby workers. Eight additional cases were identified in the only two other abattoirs using this technique. The three abattoirs that used this technique have stopped brain removal, and no new cases have been reported after 24 months of follow up. Cases compared to controls had higher median interferon-gamma (IFN γ) levels (21.7 pg/ml; vs 14.8 pg/ml, $P < 0.001$).

Discussion This novel polyradiculoneuropathy was associated with removing porcine brains with compressed air. An autoimmune mechanism is supported by higher levels of IFN γ in cases than in controls consistent with other immune mediated illnesses occurring in association with neural tissue exposure. Abattoirs should not use compressed air to remove brains and should avoid procedures that aerosolize CNS tissue. This outbreak highlights the potential for respiratory or mucosal exposure to cause an immune-mediated illness in an occupational setting.